

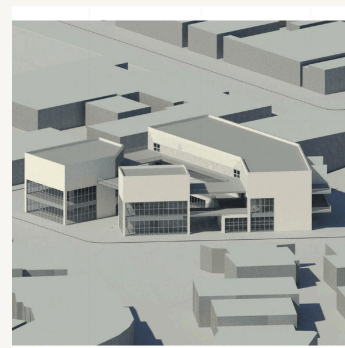
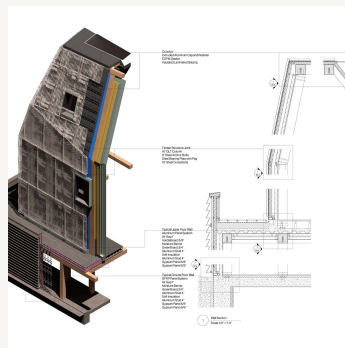
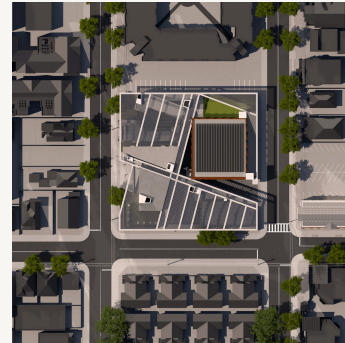
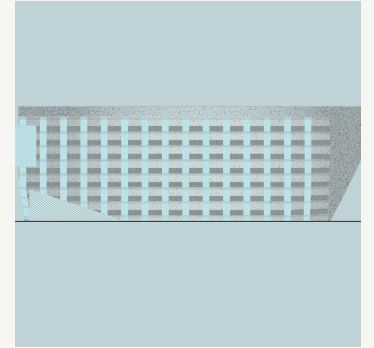
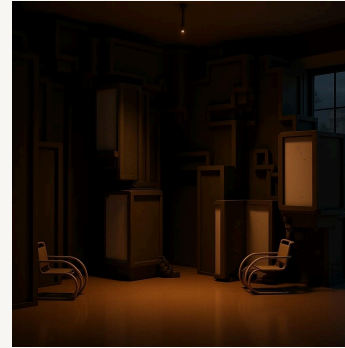
PORTFOLIO · 2023–2026

John Northrup

Designer specializing in the intersection of AI, automation, and architecture.

I build custom generative pipelines using Python, Grasshopper, and ComfyUI to solve complex spatial problems — streamlining 3D workflows through data-driven procedural modeling and advanced digital fabrication.

jcnorthrup.com



CONTENTS

01	Prototype	SP26, Prototype, 3GB Studio	03
02	Interface Architecture	3GA Studio	06
03	Performing Arts Theater in Little Tokyo	FA23, DS 1100, 1GA Studio	10
04	A Municipal Building in Van Nuys	SP24, DS 1101, 1GB Studio	16
05	LA Dance Shed	SP24, 2GA Studio, with Sledge Milan	19
06	Casa Musica	FA24, Visual Studies III, 2GA Studio	21
07	Design Documents	SP25, Design Documents	25
08	Advanced Project Delivery	SP25, Advanced Project Delivery	28
09	Details, Details	F25, Details, Details	31



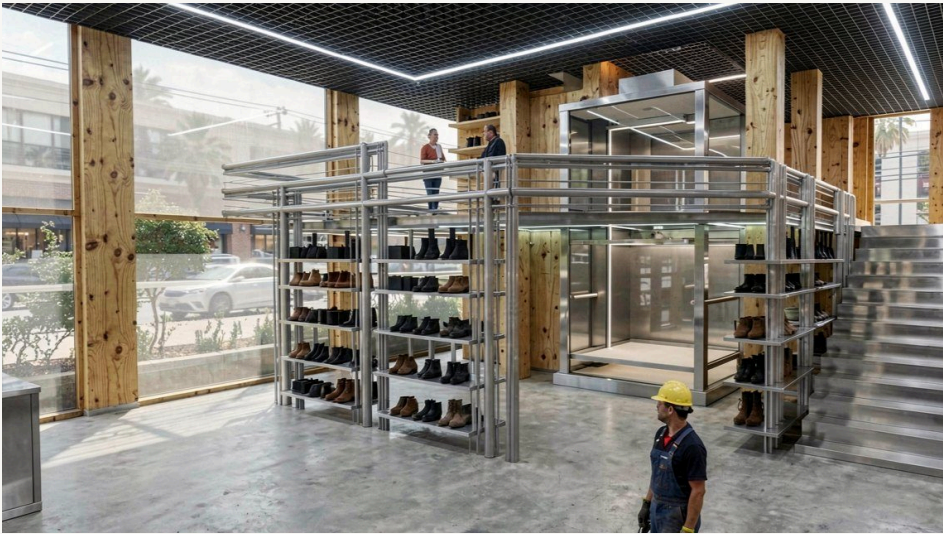
01

SP26, PROTOTYPE, 3GB STUDIO

Prototype

A generative pipeline for retail architecture and product design, built around a footwear brand's flagship storefront. Architectural iterations were produced through a custom Flux/ComfyUI pipeline, while the product line — a technical boot shown here in cutaway and colorway studies — was developed in parallel

through a Gemini -based image pipeline. The two threads feed each other: the building's industrial material palette of raw wood, steel, and concrete is carried directly into the product's own construction and display.







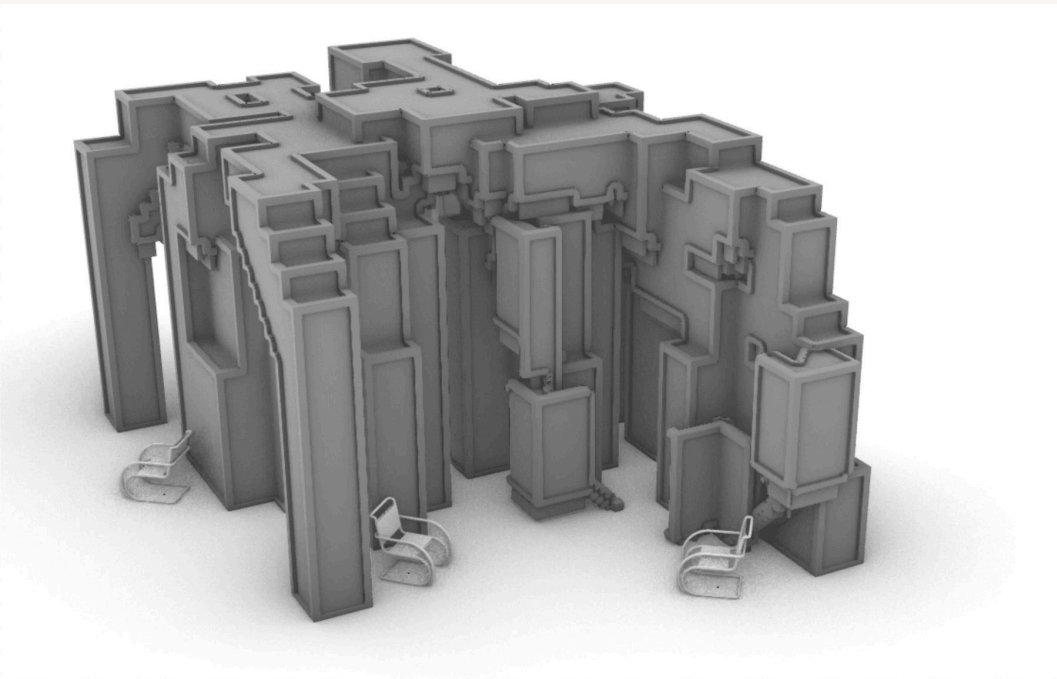
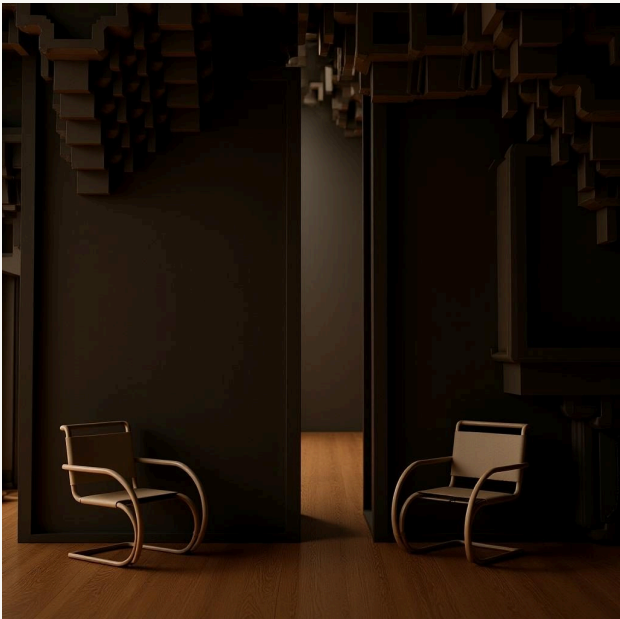
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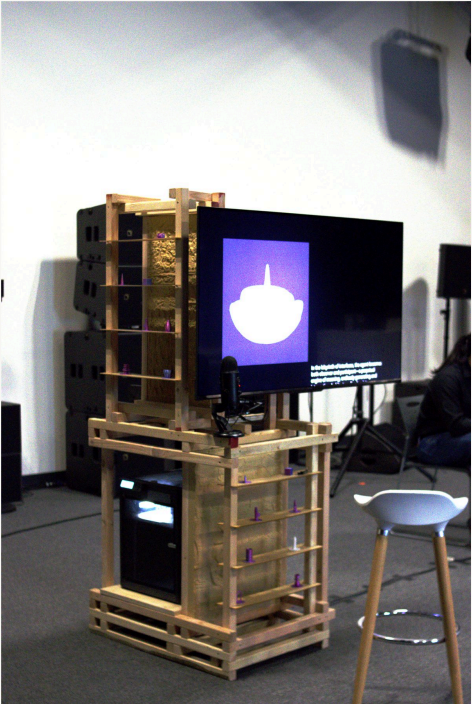
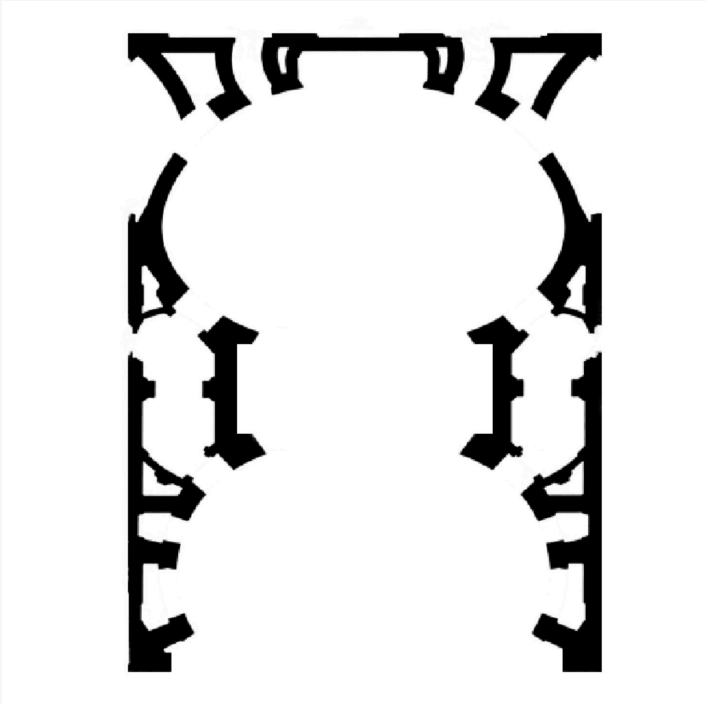
3GA STUDIO

Interface Architecture

Advisor: Casey Rehm

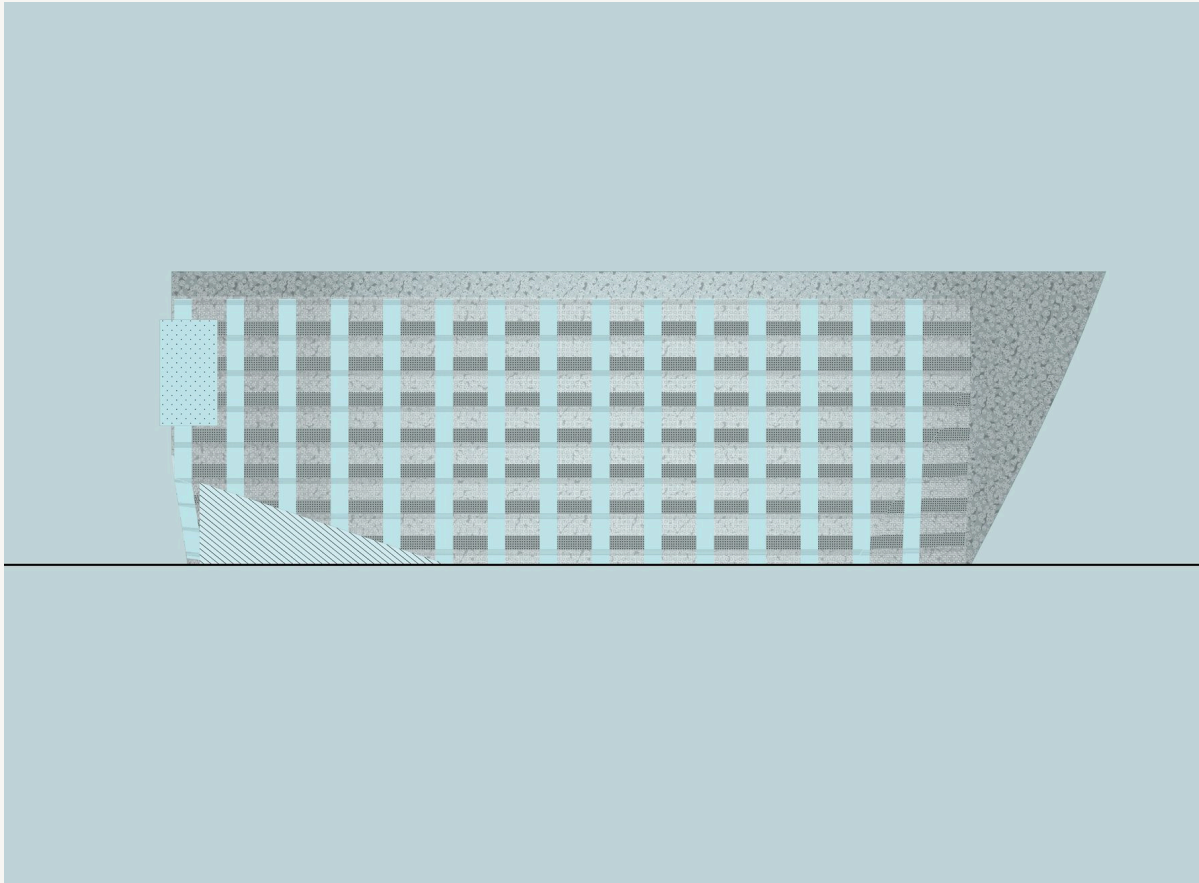
A Cabinet of Curiosities that produces memory artifacts. The installation receives audio input from the surrounding space; language becomes image through translation of an LLM to 3D object creation via API calls (Meshy). Objects are automatically optimized for 3D printing — oriented, scaled, and combined through a Python pipeline.





OBJECT STUDIES





03

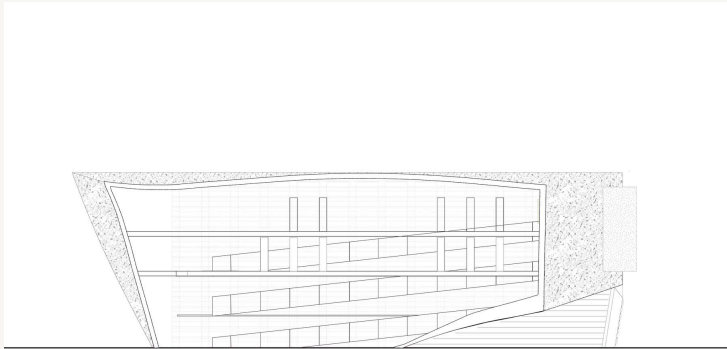
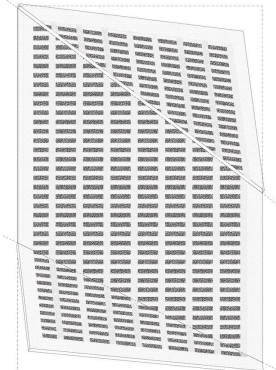
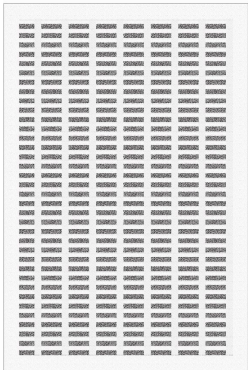
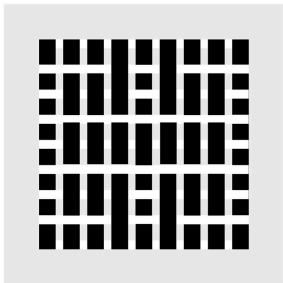
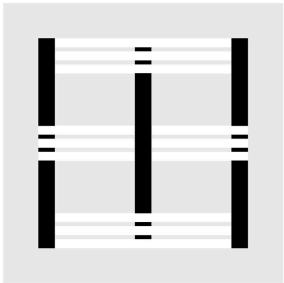
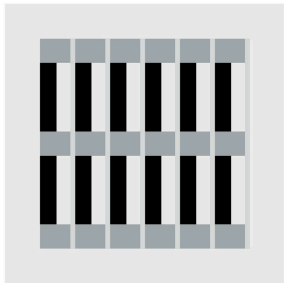
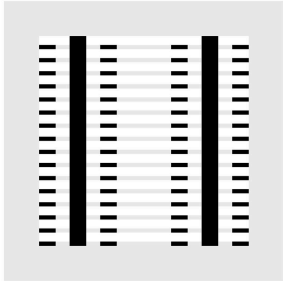
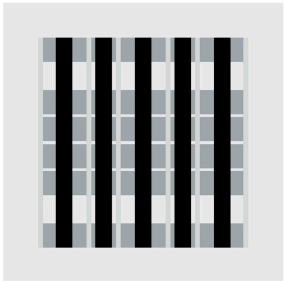
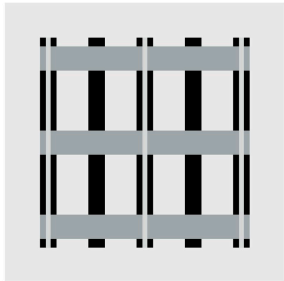
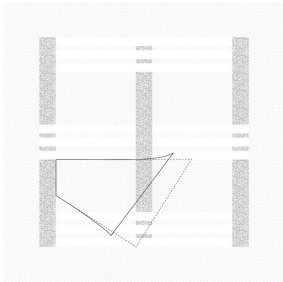
FA23, DS 1100, 1GA STUDIO

Performing Arts Theater in Little Tokyo

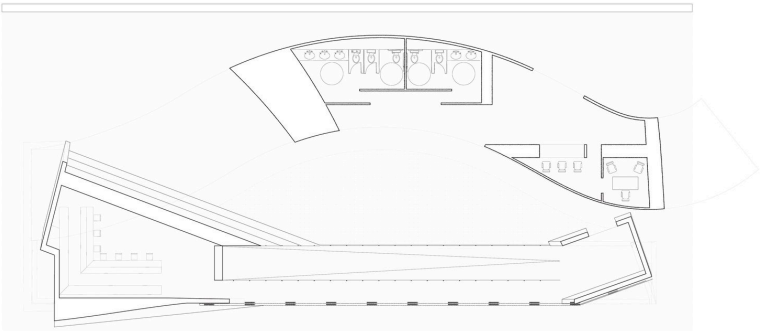
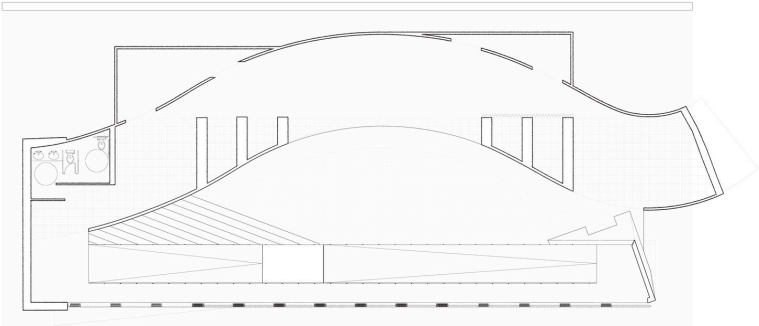
Advisor: Matthew Au

This first semester studio project stems from a study of grid systems, materiality, galleries, and theatres. The initial measurement grids were carefully constructed, using the limitations of the tape itself and the gap in named measurements vs. actual measurements. The grids operate in a way that looks at material falsities, as well as transparency. Through these measurement systems, they are then expanded into large boards. These boards are then used to produce a massing model. Ultimately realized as a performing arts theater in the Little Tokyo District, the Measurement Grid is transformed into a space that offers a place for community to gather in an unconventional theater setting where actors and patrons are active participants in the shows on display.

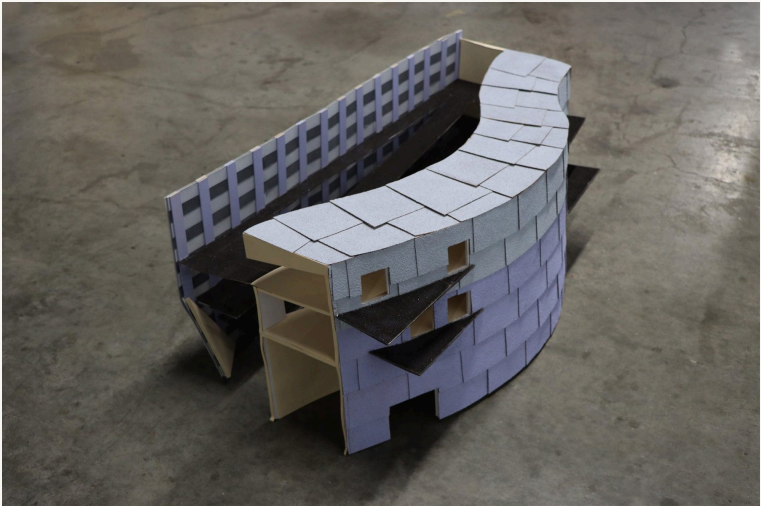
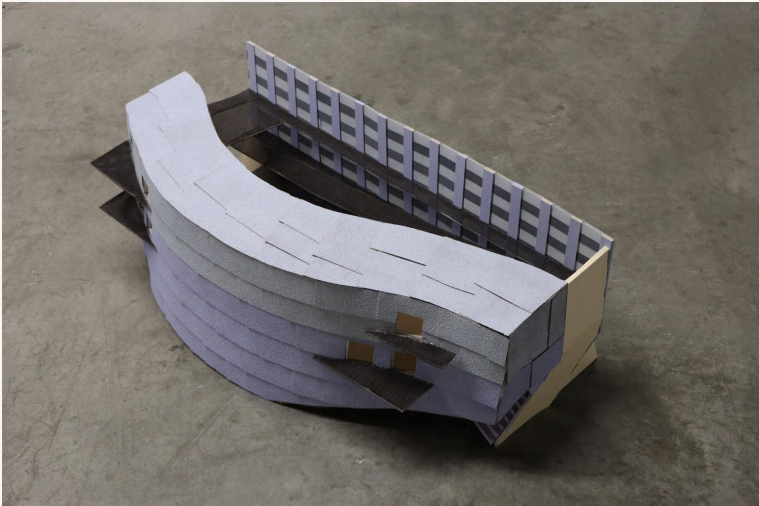
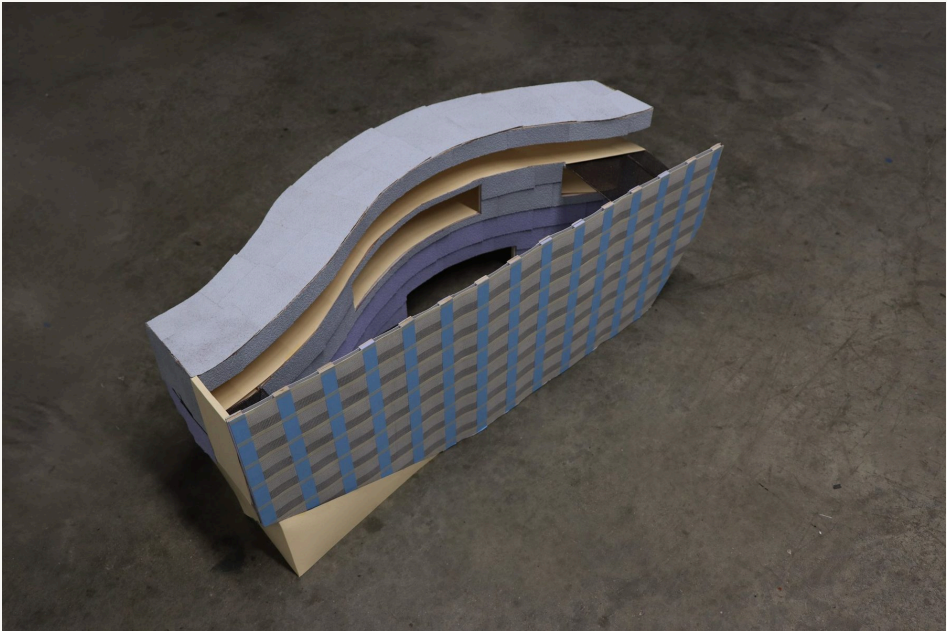
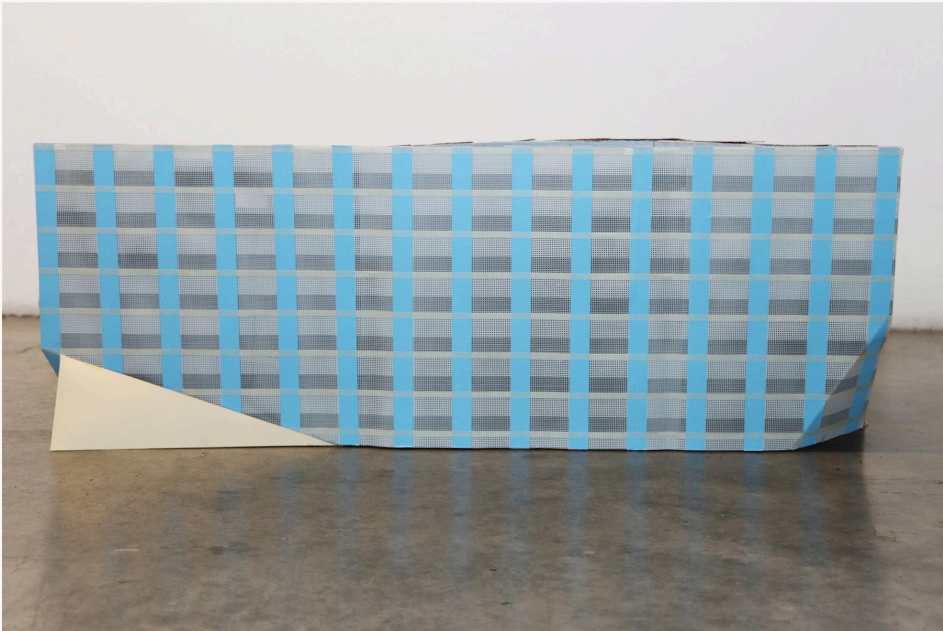
GRID STUDIES



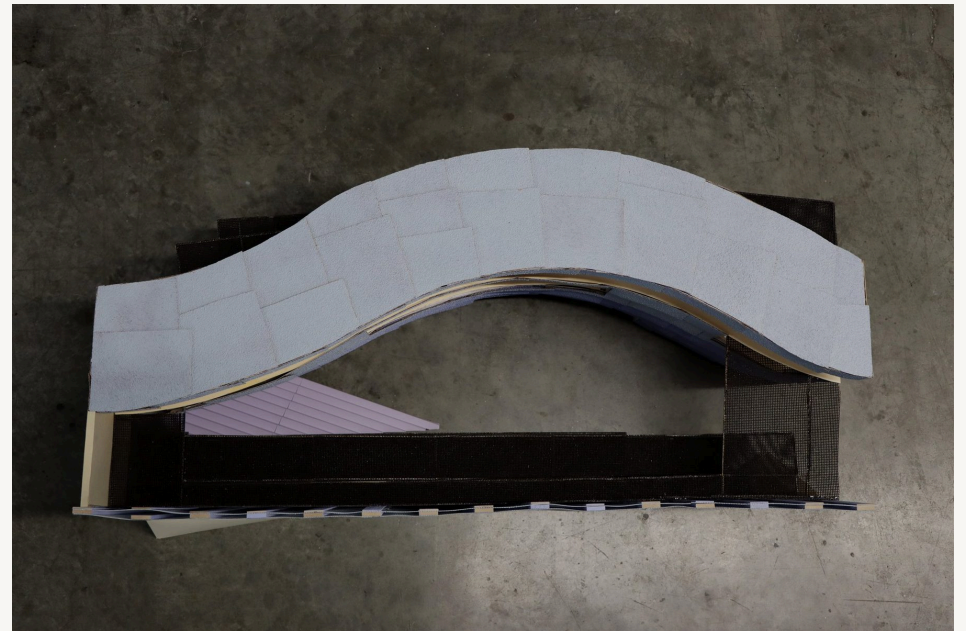
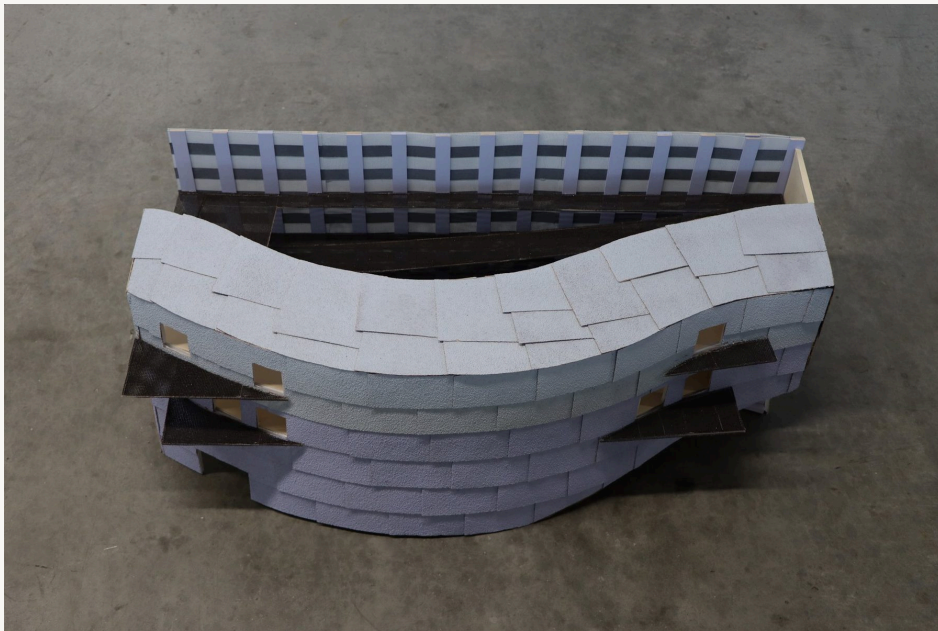
GRID STUDIES



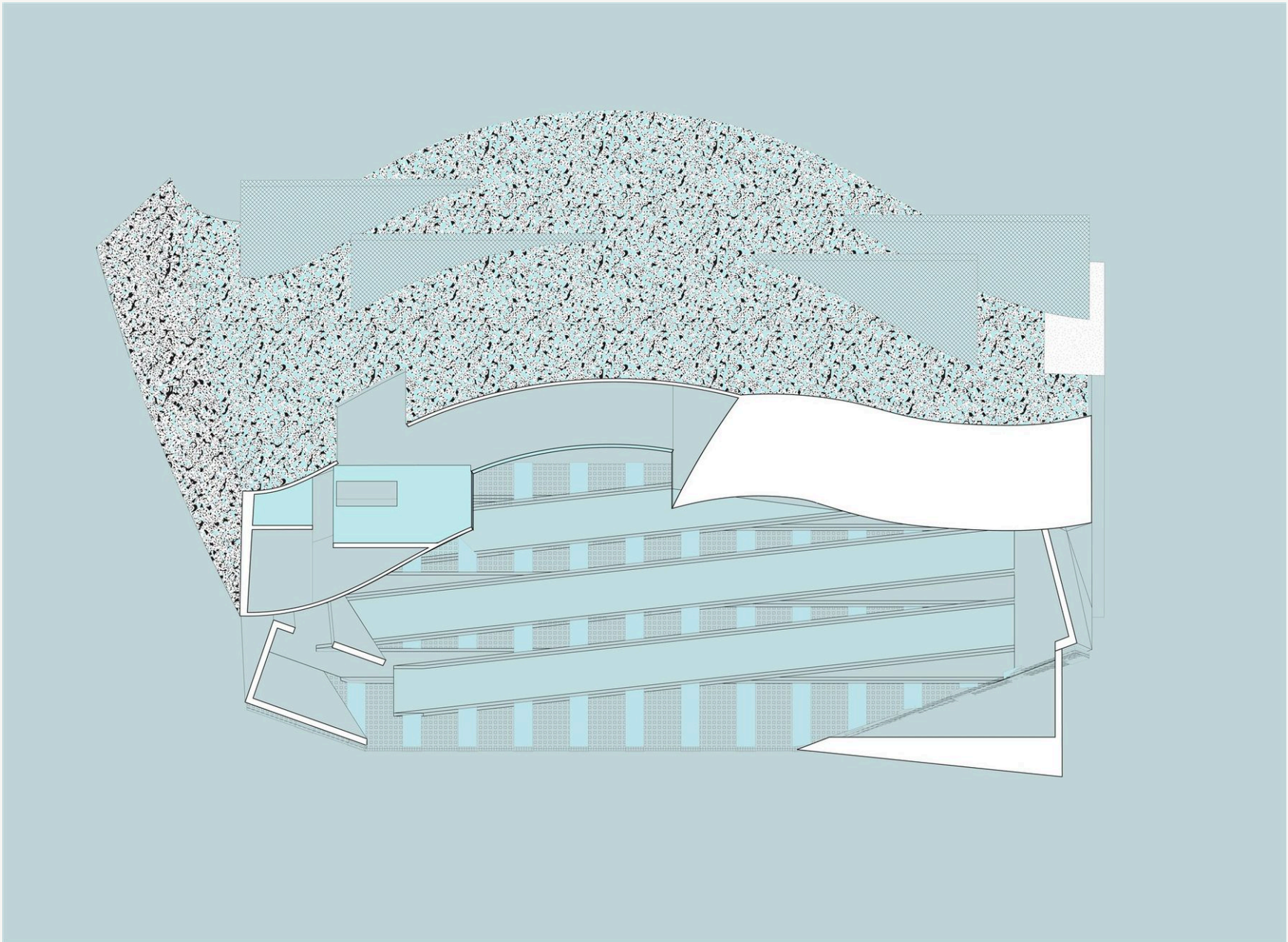
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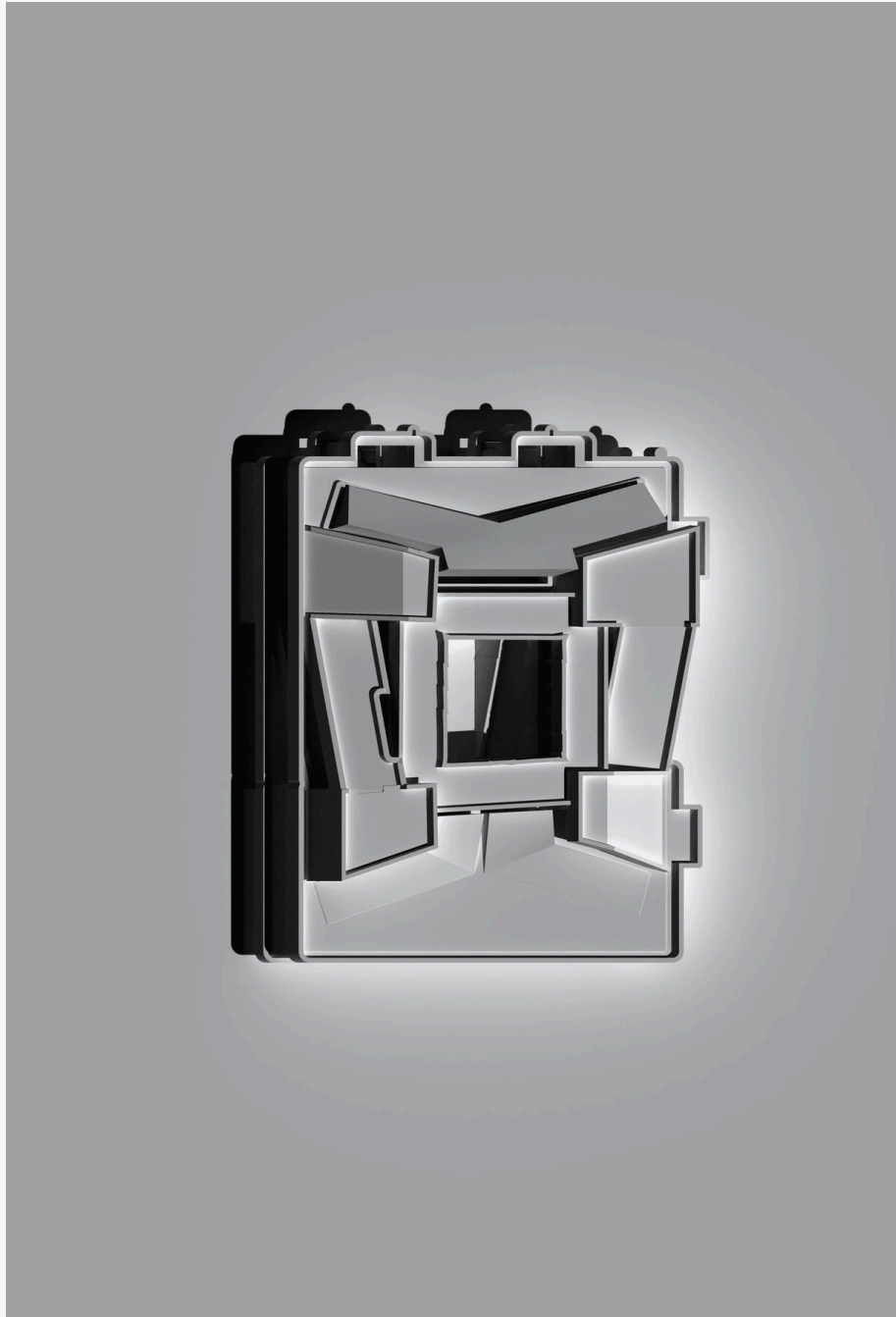


FINAL MODEL



FINAL MODEL





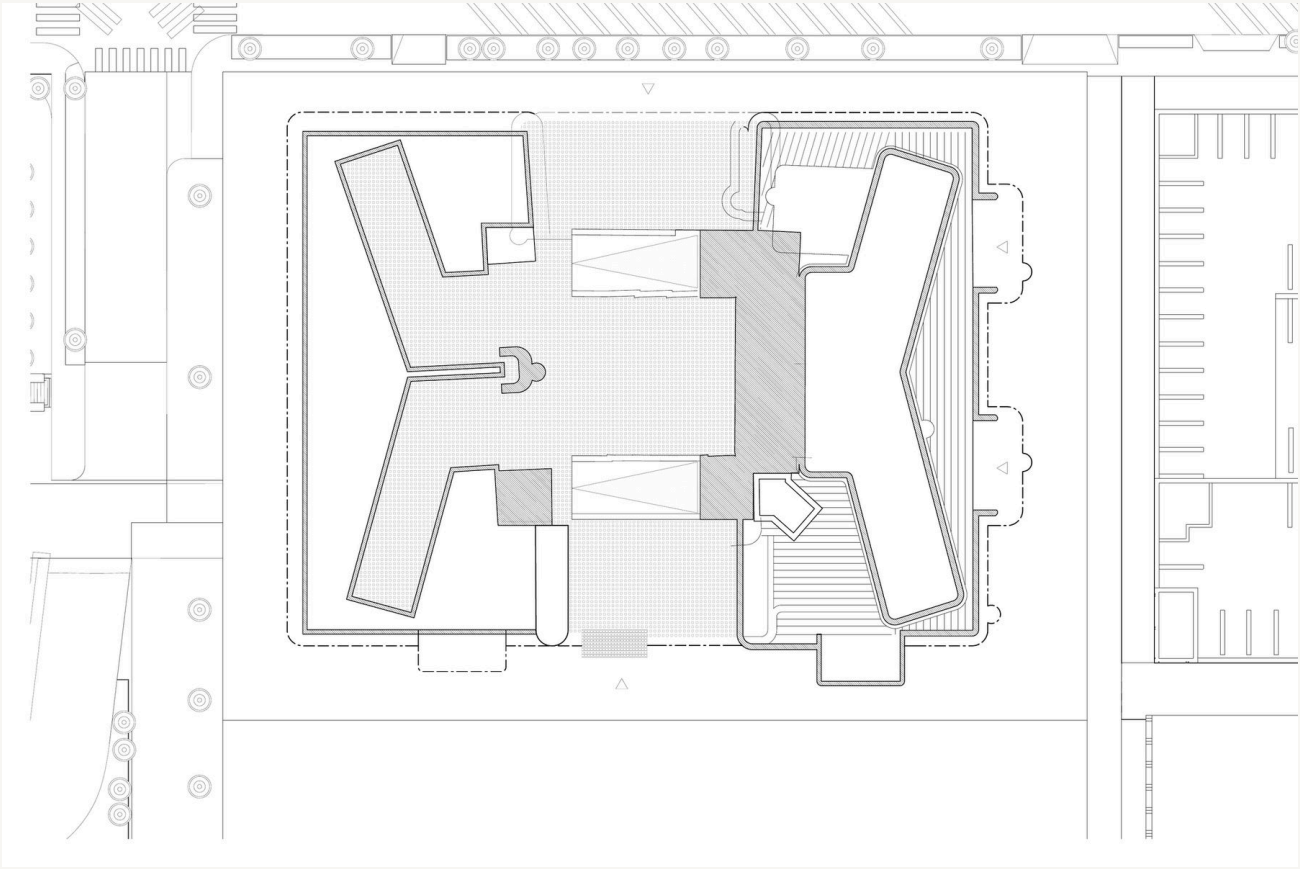
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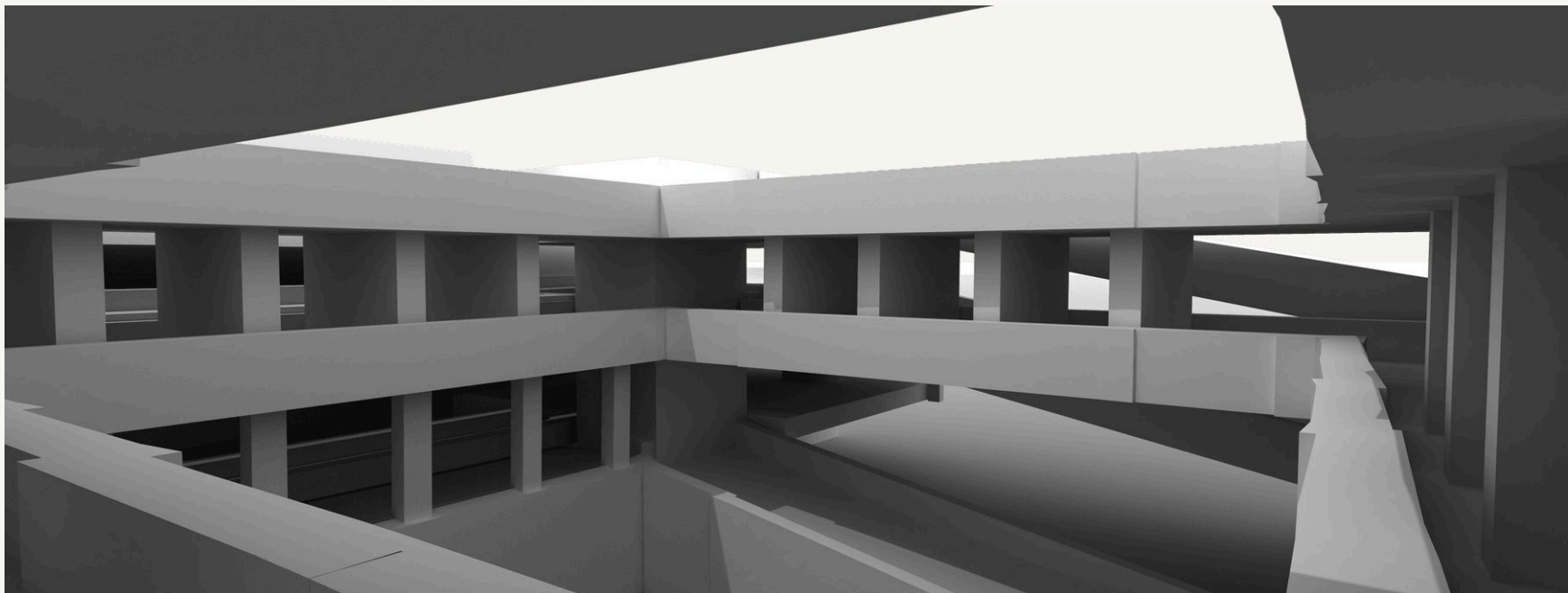
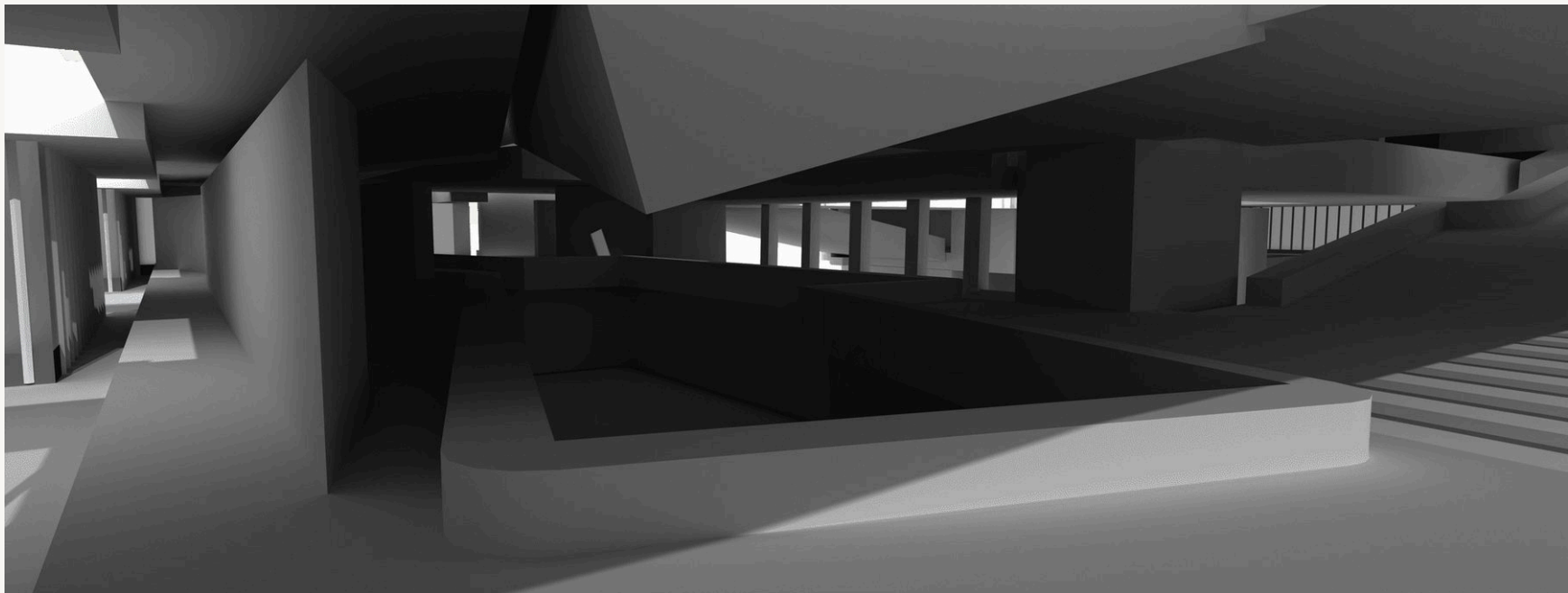
SP24, DS 1101, 1GB STUDIO

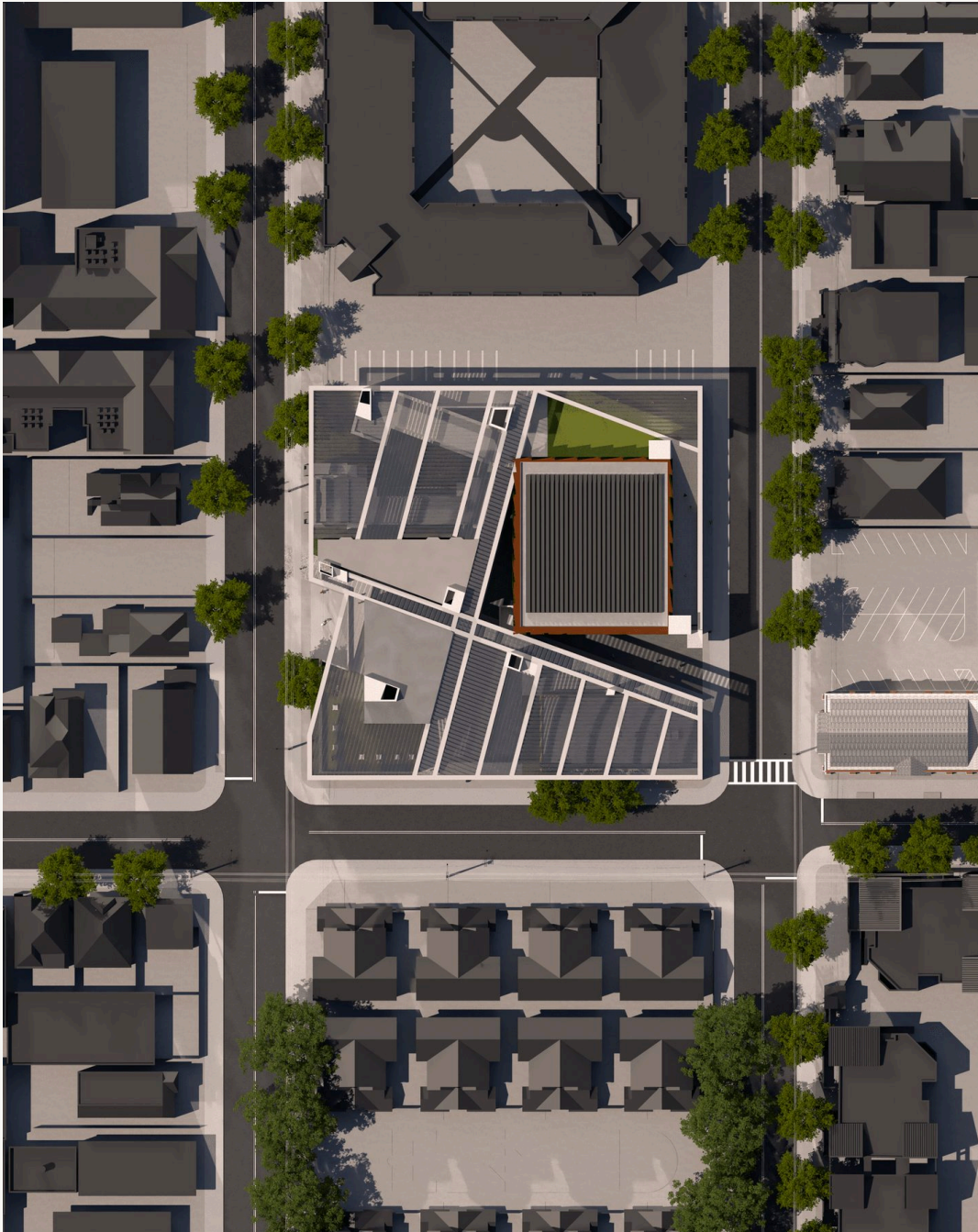
A Municipal Building in Van Nuys

Advisor: Anna Neimark

This studio course explored the unlikely pairings of parking garages and municipal buildings, producing an interaction between various spaces which often results in overlap, much like a collage of repurposed materials, the project creates a palimpsest of various conundrums. Through various studies of redaction, erasure, adaptation, annotation and material/production methodologies these works were produced. The acts of erasure and redaction are ones often violent or harmful, can also be useful mechanisms for clearing, filtering, compartmentalizing and synthesizing. Various influences were combined here, from the base studies of the 17th Century Fortress, Castillo San Marcos, the Faculdade de Arquitetura, Daniel Libeskind's micromegas. The contrasting components become a study of composed disorganization, collage, and porosity.





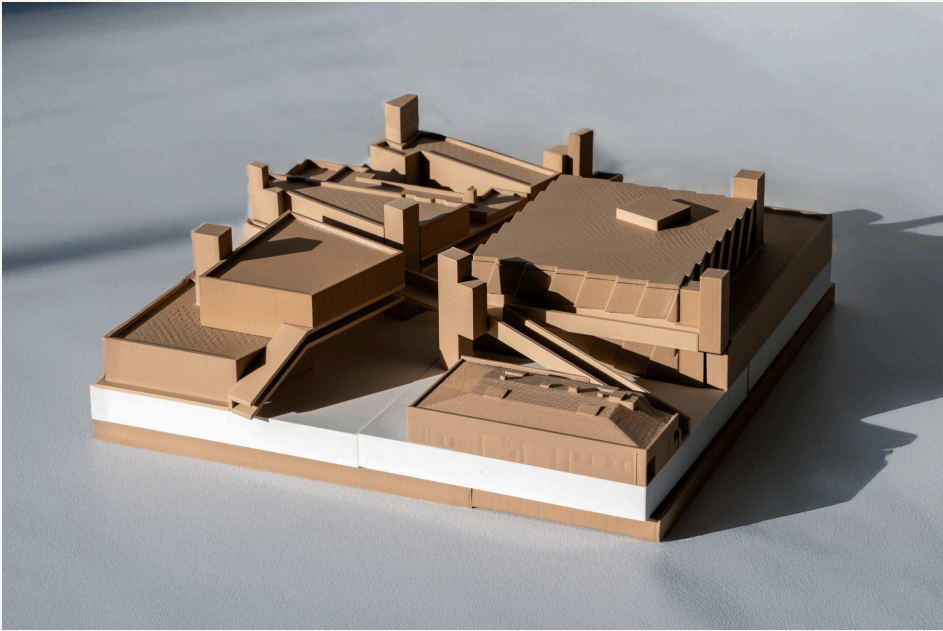
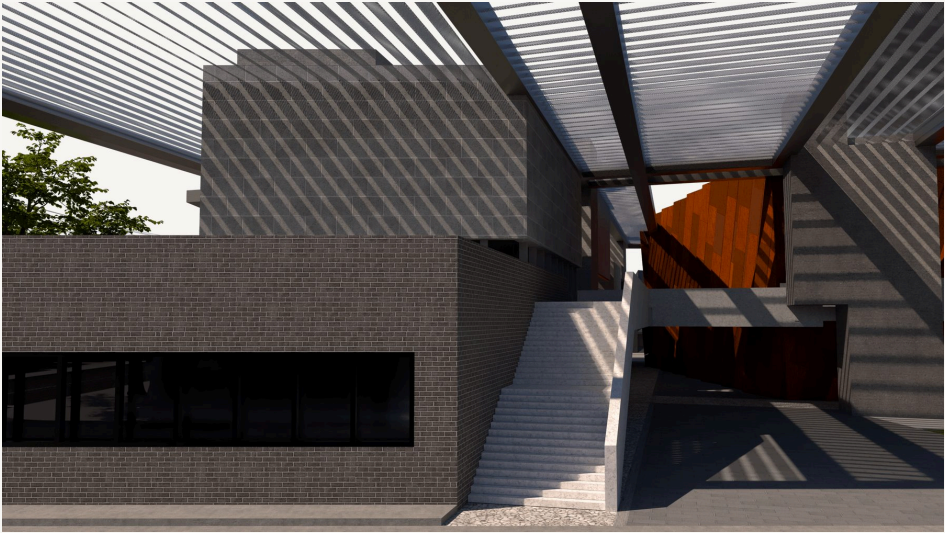


05

SP24, 2GA STUDIO, WITH SLEDGE MILAN

LA Dance Shed

This project for the Los Angeles Dance Project (LADP) seeks to bring the community of Pico-Union together on a campus that serves and connects the community. Located directly across from the Pico-Union Project, the site was surgically opened up in order to connect via sitelines and circulation to this important community asset. A unique goal among this project was to preserve many of the existing structures on the site and repurposing the materials from buildings that were to be removed. With a raised circulatory system, visitors are encouraged to meander throughout the site, engaging with it from within the in-between spaces. The roof serves as not only shade for warm summer days, but also acts as an organizing device for the discreet elements of the site, bringing them together in a collective, just as is intended with the surrounding community.





06

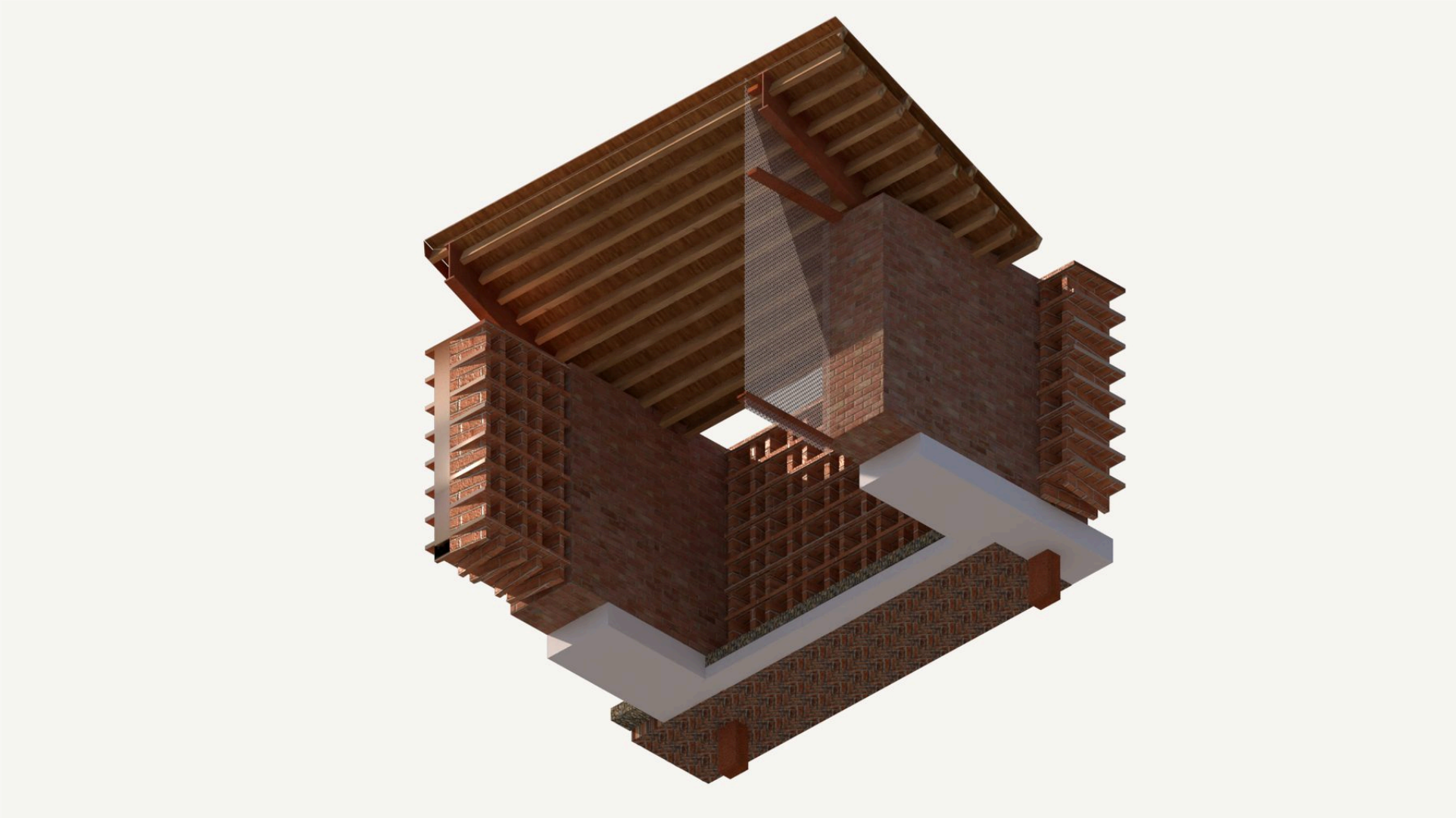
FA24, VISUAL STUDIES III, 2GA STUDIO

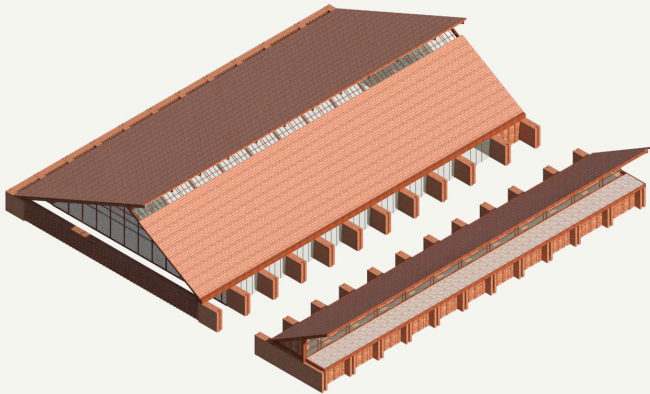
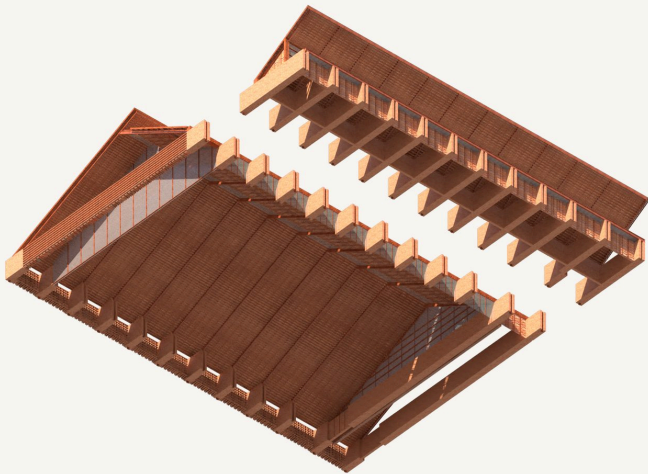
Casa Musica

Advisor: Marcelo Spina

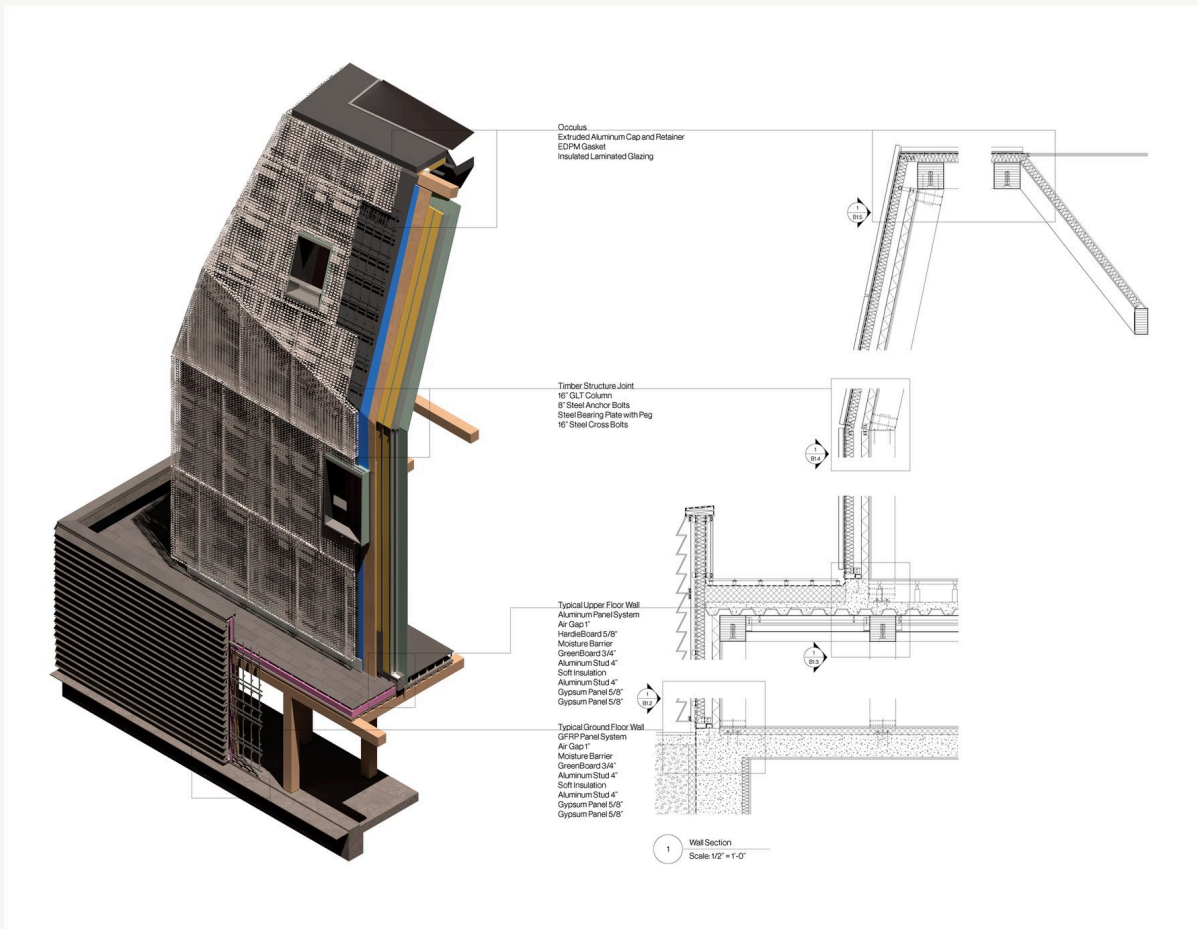
Casa Musica sits within an active desert quarry, its long gabled roofs and sawtooth wings pitched to echo the surrounding excavated terrain. A brick base grounds the structure while a heavy timber roof structure spans overhead, its gable end fully glazed to frame the quarry and mountains beyond. Inside, the

room is deliberately unfinished — raw stone brought in from the site sits alongside cast furniture, blurring the line between what was quarried and what was built. The result is a communal gathering space for music and performance that reads as much as an extension of the landscape as a building set within it.









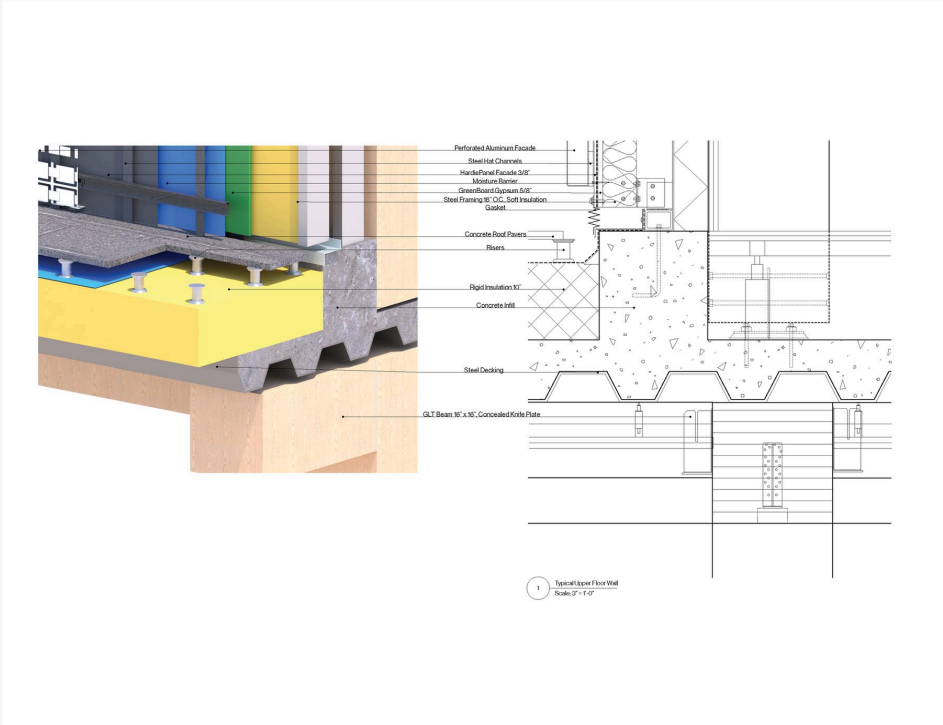
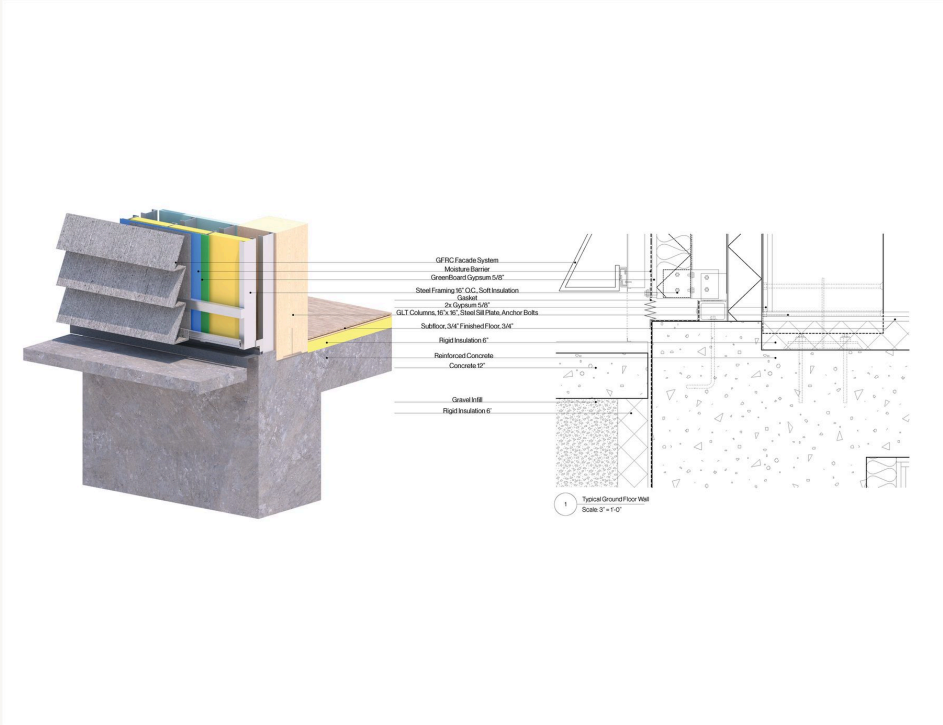
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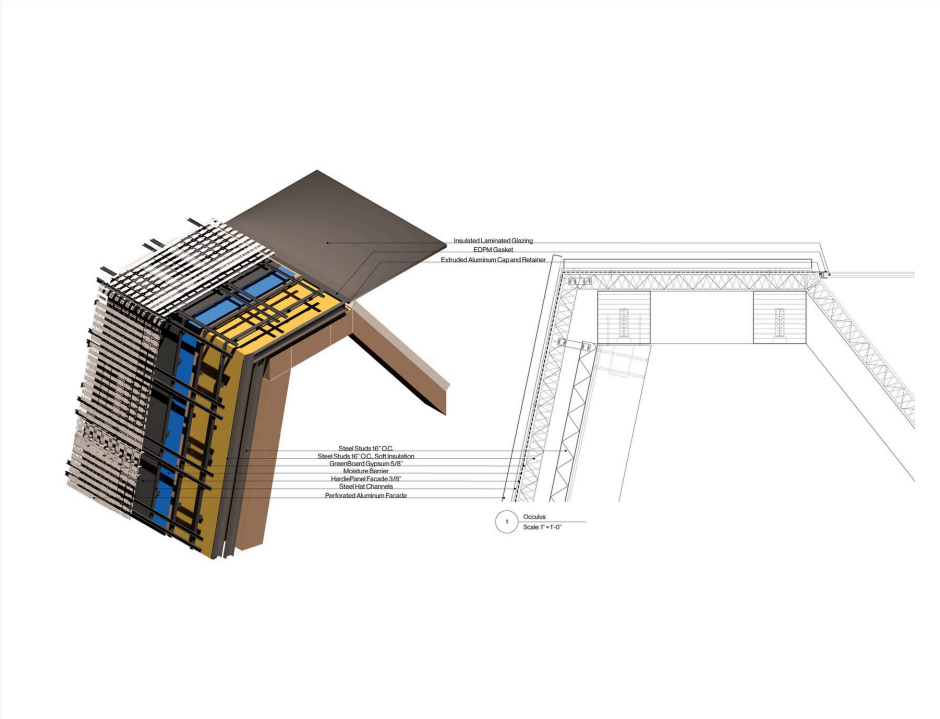
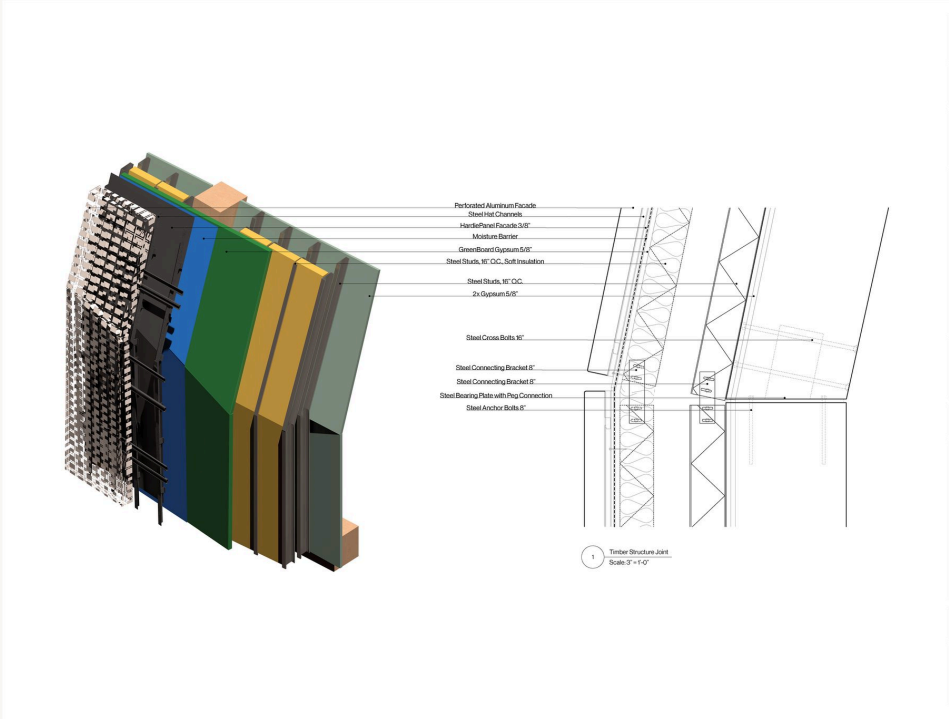
SP25, DESIGN DOCUMENTS

Design Documents

Advisor: Herwig Baumgartner

This course studies the implementation of design, technology, and an understanding of building systems. Students are tasked with preparing Design Document drawings from a previous year's studio project and are asked to bring it to life with real - world materials, budgets, and environmental factors.







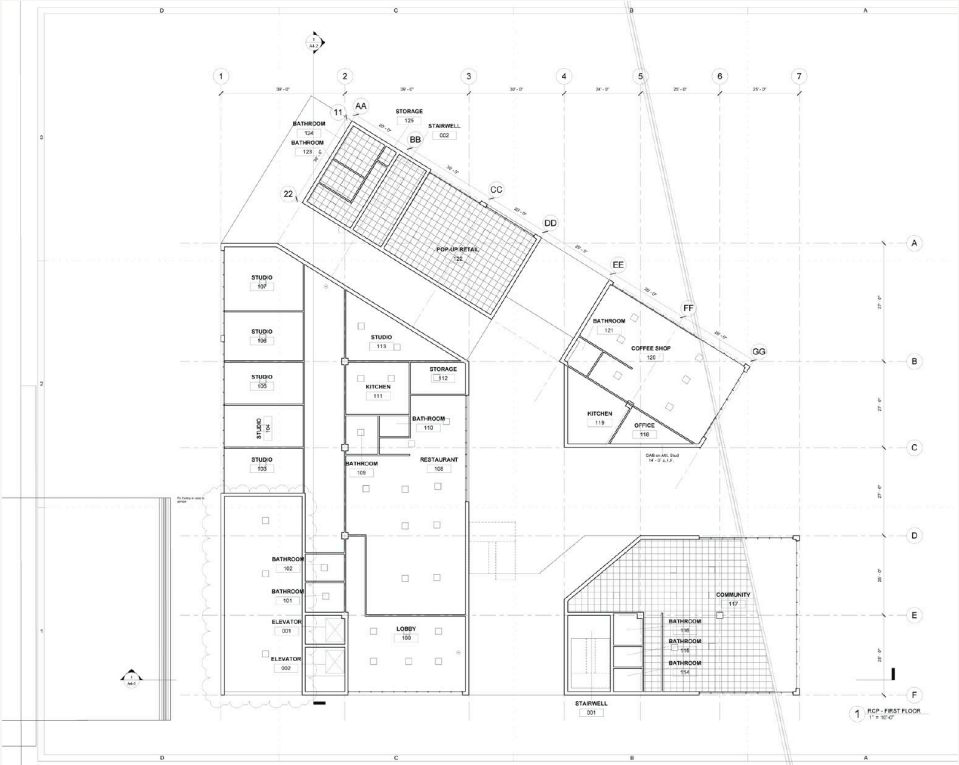
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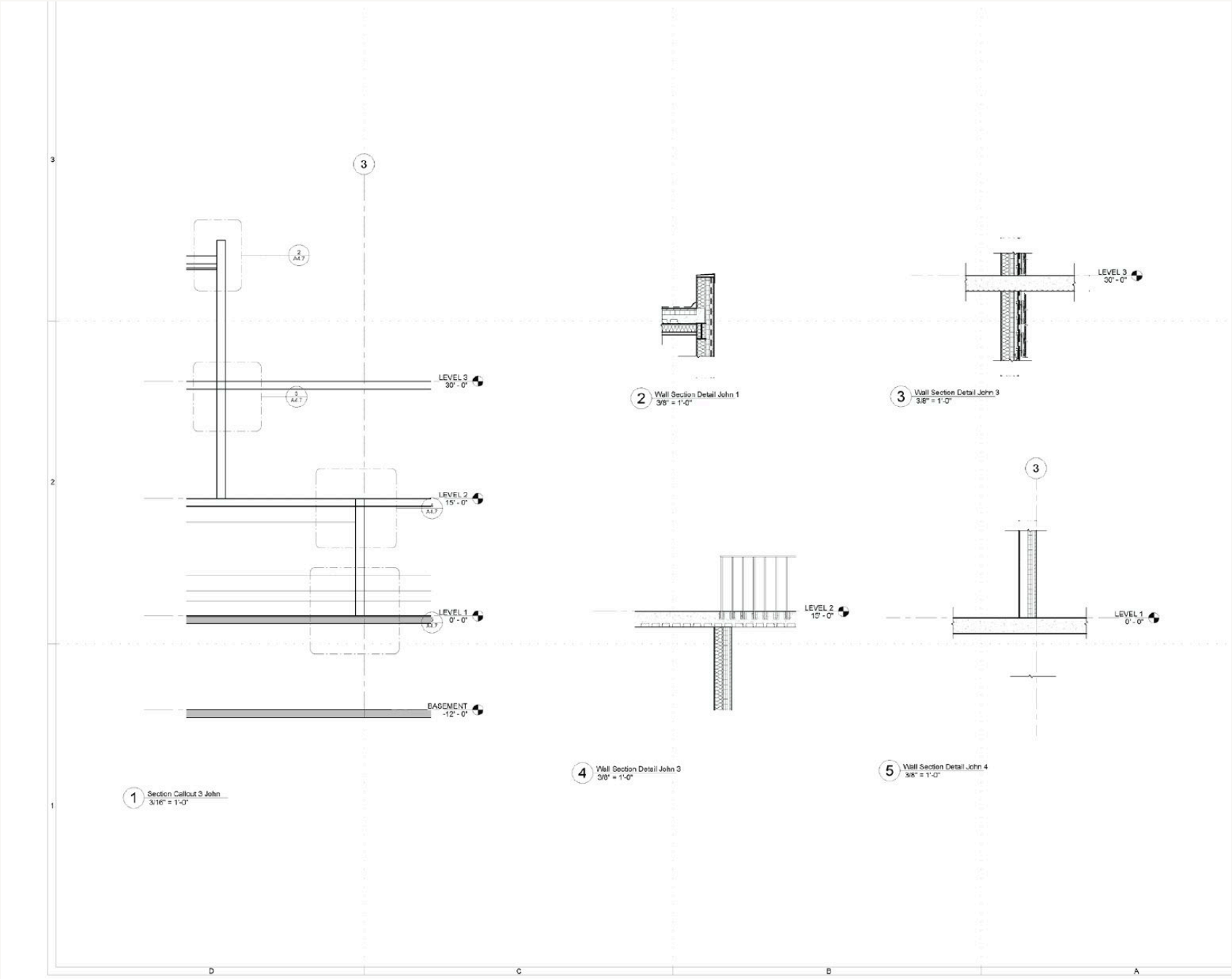
SP25, ADVANCED PROJECT DELIVERY

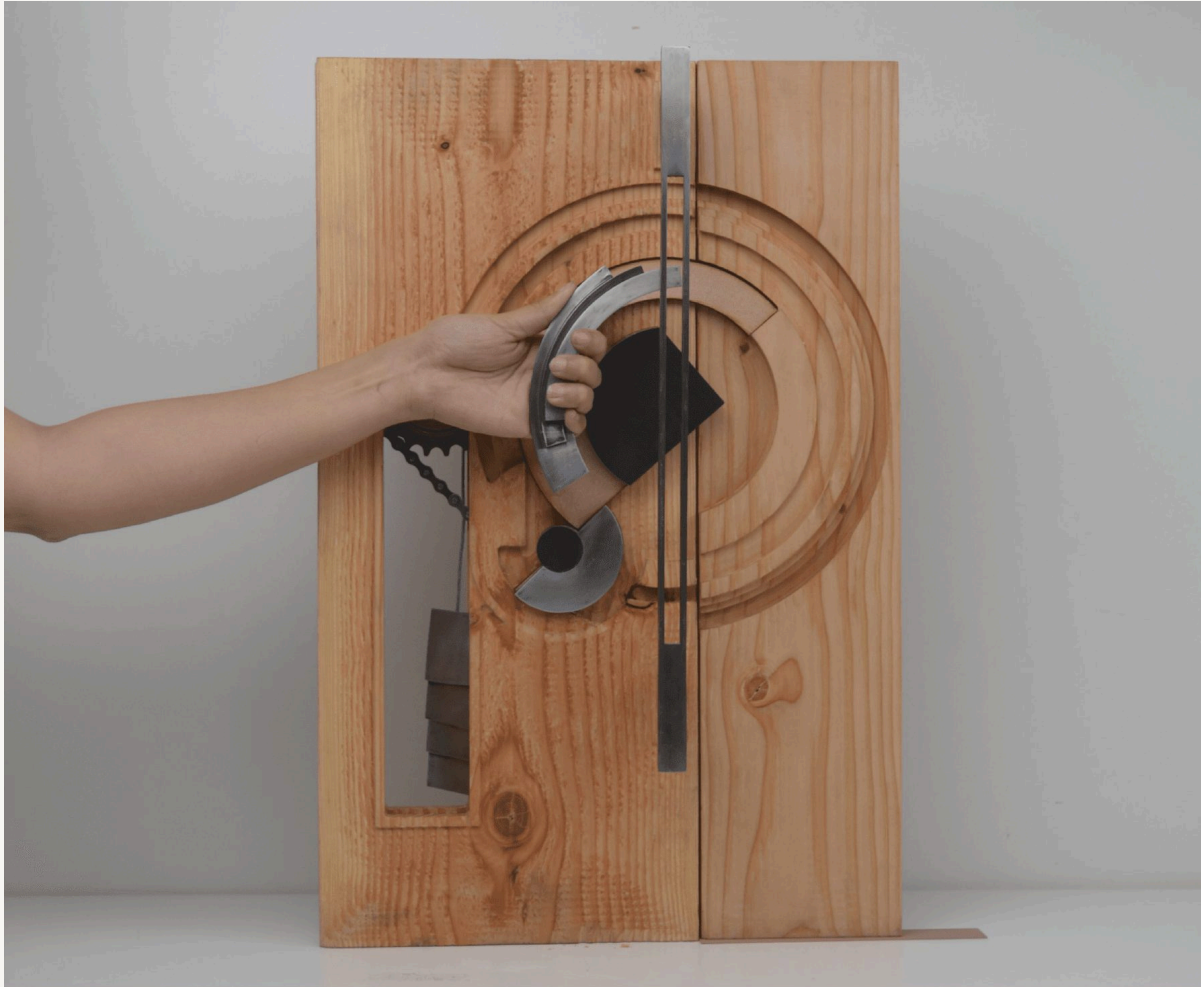
Advanced Project Delivery

Advisor: Pavel Getov and Karenza Harris

A Construction Documents course focused on taking a design through to buildable detail — resolving material assemblies, structural connections, and building systems into a coordinated drawing set. Site plans, reflected ceiling plans, and wall section sheets were developed to real-world standards of dimension, specification, and constructability.







09

F25, DETAILS, DETAILS

Details, Details

Advisor: Dwayne Oyler

A doorlock mechanism inspired by Frank Lloyd Wright's detail work, specifically the door mechanisms in the Hollyhock House.

**All work made with partners Chia-Chen Li and Baizhen Yu*

